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Replication

DNA is the carrier of genetic information. Before a cell divides, DNA must be precisely copied, or "replicated," so that each of the two daughter cells can inherit a complete genome, the full set of genes present in the organism. In **eukaryotes**, the DNA molecules that make up the genome are packaged with proteins into chromosomes, each of which contains a single linear DNA molecule. Eukaryotic chromosomes are found in a special compartment called the cell nucleus. The genomes of bacterial cells (**prokaryotes**), which lack a nucleus, are typically circular DNA molecules that associate with special structures in the cell membrane. Despite the hundreds of millions of years of evolutionary history separating eukaryotes and prokaryotes, the features of the replication process have been highly conserved between them.

Overview

The DNAs that make up the genomes of bacteria and eukaryotic cells are double-stranded molecules in which each strand is composed of subunits called nucleotides. DNA nucleotides have a direction, in the same way that an arrow has a head and a tail. In DNA strands, the head is the 3' ("three prime") end of the strand, and the tail is the 5' ("five prime") end. As a result, each strand also has a direction, whose ends are referred to as the 3' and 5' ends. The two strands of DNA run in opposite directions, and are wound around each other in a double helix, with the strands held together by **hydrogen bonds** between paired bases of the nucleotides (A pairs with T, and G pairs with C).

During the process of DNA replication, the strands are unwound by an enzyme called DNA helicase, and a new strand of DNA is synthesized on each of the old (**template**) strands by an enzyme called DNA polymerase, which joins incoming nucleotides together in a sequence that is determined by the sequence of nucleotides present in the template strand. DNA replication is said to be semiconservative because each of the two identical daughter molecules contains one of the two parental template strands paired with a new strand. Prokaryotic replication can take as little as twenty minutes, while replication in eukaryotes takes considerably longer, approximately eight hours in mammals.

Initiation of DNA Replication

DNA replication begins (initiates) at special sites called origins of DNA replication. Eukaryotic DNAs each contain multiple replication origins,

eukaryotes organisms with cells possessing a nucleus

prokaryotes single-celled organisms without a nucleus

hydrogen bonds weak bonds between the H of one molecule or group and a nitrogen or oxygen of another

template a master copy

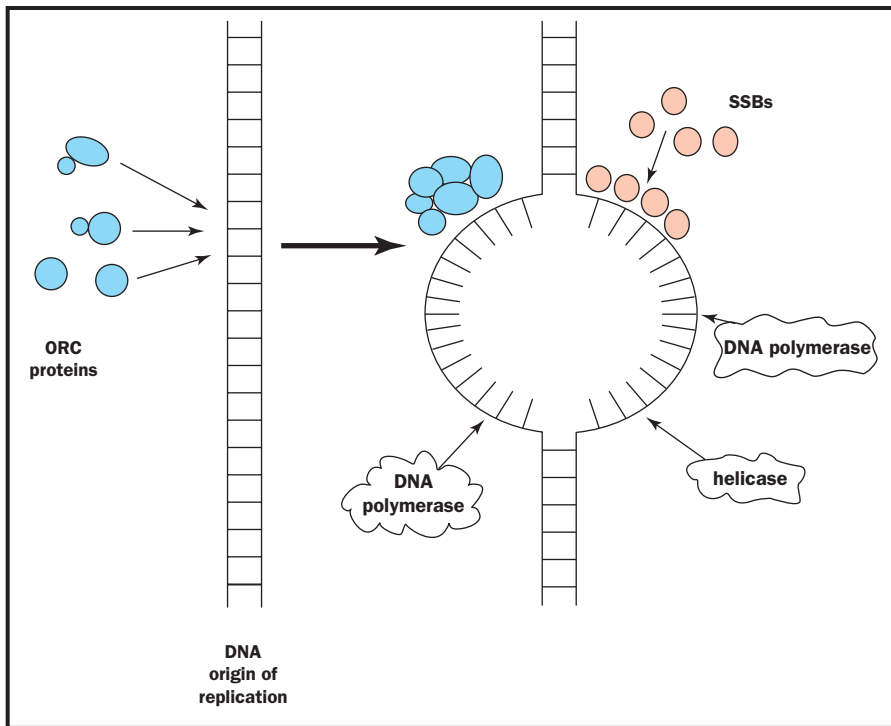


Figure 1. DNA replication begins when ORC proteins attach to an origin of replication, of which there are thousands throughout the genome. Helicase unwinds the double helix, and single-strand binding proteins stabilize it, while it is copied by DNA polymerases.

spaced at intervals of approximately 100,000 base pairs (100 kilobase pairs, or 100 kb) along the length of the DNA. There are 6 billion **base pairs** in the human genome, located on forty-six chromosomes, and so each chromosome will have many origins of replication. Prokaryotic chromosomes typically have a single replication origin.

Replication origins are composed of special sequences of DNA that are recognized by replication initiator proteins, which bind to the origin sequences and then help to assemble other proteins required for DNA replication at these sites. The eukaryotic replication initiator protein is a complex containing six different subunits called the origin recognition complex (ORC). The bacterial replication initiator protein is called the *dnaA* protein. The timing of DNA replication is regulated by controlling the assembly of complexes at replication origins.

The distinct steps in the initiation of replication are understood better in bacteria than in eukaryotes, but several key steps are common to both. The first step is a change in the conformation of the initiator protein, which causes limited “melting” (that is, the separation of the two strands) of the double-stranded DNA next to the initiator binding site, thus exposing single-stranded regions of the template (Figure 1). Two more proteins, DNA helicase and DNA primase, then join the complex. Replication initiation is triggered by the activation of the helicase and primase, and the subsequent recruitment of DNA polymerase. In prokaryotes, the particular form of the enzyme is called DNA polymerase III. Other proteins are also recruited, each of whose functions are discussed below.

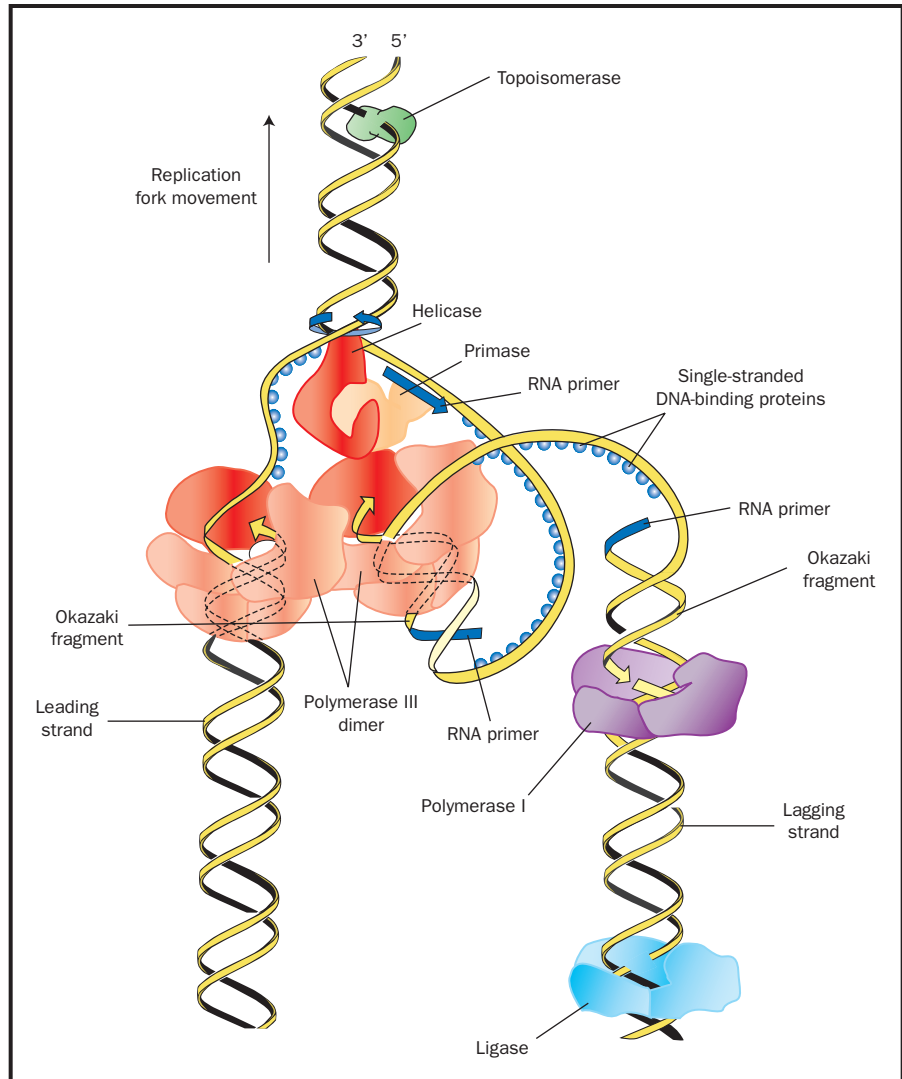
The Replication Fork

The separation of the two template strands and the synthesis of new daughter DNA molecules creates a moving “replication fork” (Figure 2), in which

base pairs nucleotides (either DNA or RNA) linked by weak bonds

Inhibitors of viral helicase-primase enzymes are being tested as a new treatment for herpes virus infection.

Figure 2. Model of a bacterial replication fork.



double-stranded DNA is continually unwound and copied. The unwinding of DNA poses special problems, which can be visualized by imagining pulling apart two pieces of string that are tightly wound around each other. The pulling apart requires energy; the strands tend to rewind if not held apart; and the region ahead of the separated strands becomes even more tightly twisted.

Proteins at the replication fork address each of these problems. DNA polymerases are not able to unwind double-stranded DNA, which requires energy to break the hydrogen bonds between the bases that hold the strands together. This task is accomplished by the enzyme DNA helicase, which uses the energy in **ATP** to unwind the template DNA at the replication fork. The single strands are then bound by a single-strand binding protein (called SSB in bacteria and RPA in eukaryotes), which prevents the strands from reassociating to form double-stranded DNA. Unwinding the DNA at the replication fork causes the DNA ahead of the fork to rotate and become twisted on itself. To prevent this from happening, an enzyme called DNA gyrase (in bacteria) or topoisomerase (in eukaryotes) moves ahead of the

ATP adenosine triphosphate, a high-energy compound used to power cell processes

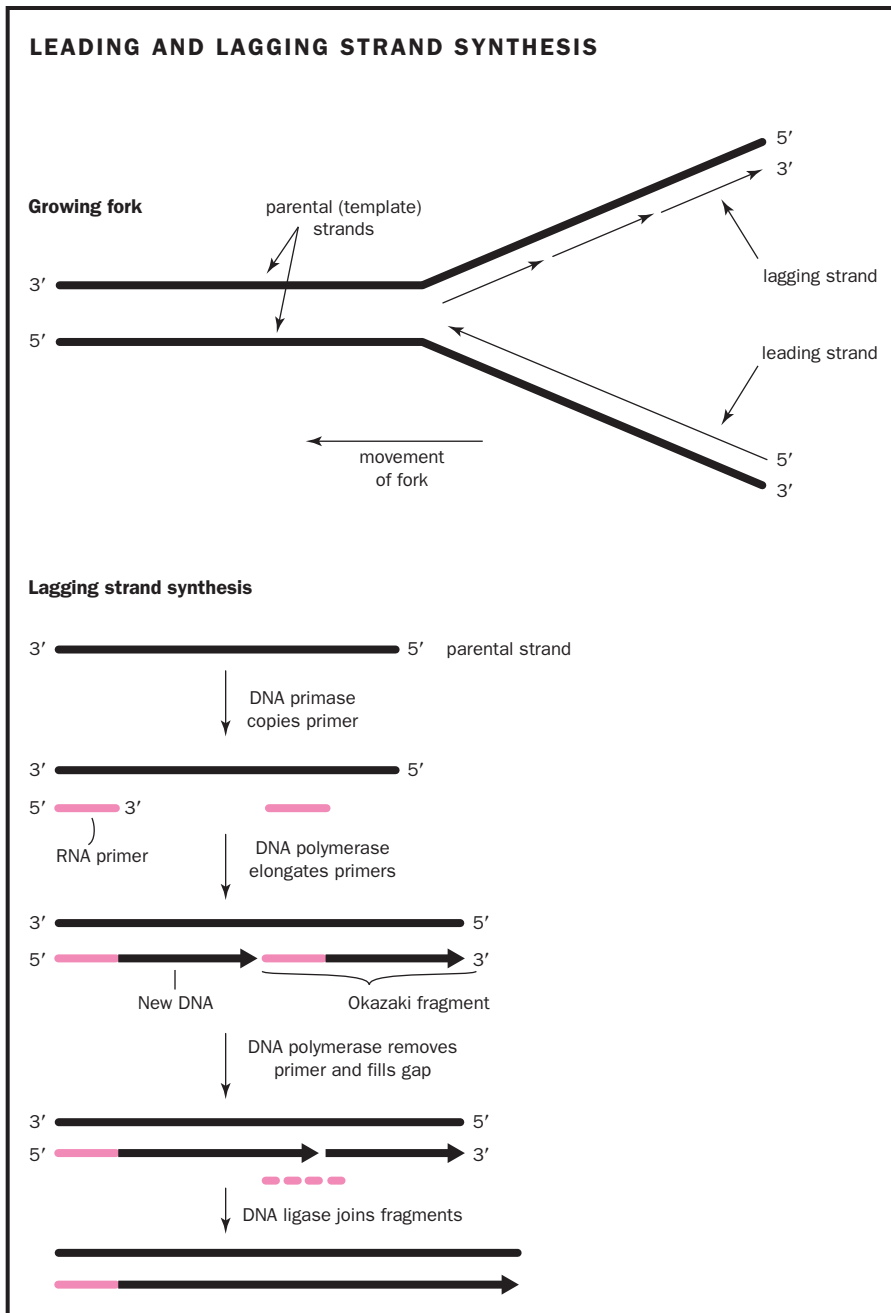


Figure 3. Leading and lagging strand synthesis. Adapted from Lodish, 1999.

replication fork, breaking, swiveling, and rejoining the double helix to relieve the strain.

Leading Strands and Lagging Strands

The coordinated synthesis of the two daughter strands posed an important problem in DNA replication. The two parental strands of DNA run in opposite directions, one from the 5' to the 3' end, and the other from the 3' to the 5' end. However, all known DNA polymerases catalyze DNA synthesis in only one direction, from the 5' to the 3' end, adding nucleotides only to the 3' end of the growing chain. The daughter strands, if they were both synthesized continuously, would have to be synthesized in opposite direc-

tions, but this is known not to occur. How, then, can the other strand be synthesized?

The resolution of the problem was provided by the demonstration that only one of the two daughter strands, called the leading strand, is synthesized continuously in the overall direction of fork movement, from the 5' to the 3' end (see Figure 3). The second daughter strand, called the lagging strand, is made discontinuously in small segments, called Okazaki fragments in honor of their discoverer. Each Okazaki fragment is made in the 5' to 3' direction, by a DNA polymerase whose direction of synthesis is backwards compared to the overall direction of fork movement. These fragments are then joined together by an enzyme called DNA ligase.

The Need for Primers

Another property of DNA polymerase poses a second problem in understanding replication. DNA polymerases are unable to initiate synthesis of a new DNA strand from scratch; they can only add nucleotides to the 3' end of an existing strand, which can be either DNA or RNA. Thus, the synthesis of each strand must be started (primed) by some other enzyme.

The priming problem is solved by a specialized RNA polymerase, called DNA primase, which synthesizes a short (3 to 10 nucleotides) RNA primer strand that DNA polymerase extends. On the leading strand, only one small primer is required at the very beginning. On the lagging strand, however, each Okazaki fragment requires a separate primer.

Before Okazaki fragments can be linked together to form a continuous lagging strand, the RNA primers must be removed and replaced with DNA. In bacteria, this processing is accomplished by the combined action of RNase H and DNA polymerase I. RNase H is a **ribonuclease** that degrades RNA molecules in RNA/DNA double helices. In addition to its polymerase activity, DNA polymerase I is a 5'-to-3' nuclease, so it too can degrade RNA primers. After the RNA primer is removed and the gap is filled in with the correct DNA, DNA ligase seals the nick between the two Okazaki fragments, making a continuous lagging strand.

ribonuclease enzyme
that cuts RNA

DNA Polymerase

The two molecules of DNA polymerase used for the synthesis of both leading and lagging strands in bacteria are both DNA polymerase III. They are actually tethered together at the fork by one of the subunits of the protein, keeping their progress tightly coordinated. Many of the other players involved are also linked, so that the entire complex functions as a large molecular replicating machine.

DNA polymerase III has several special properties that make it suitable for its job. Replication of the leading strand of a bacterial chromosome requires the synthesis of a DNA strand several million bases in length. To prevent the DNA polymerase from “falling off” the template strand during this process, the polymerase has a ring-shaped clamp that encircles and slides along the DNA strand that is being replicated, holding the polymerase in place. This sliding clamp has to be opened like a bracelet in order to be loaded onto the DNA, and the polymerase also contains a special clamp loader that does this job.

A second important property of DNA polymerase III is that it is highly accurate. Any mistakes made in incorporating individual nucleotides cause **mutations**, which are changes in the DNA sequence. These mutations can be harmful to the organism. The accuracy of the DNA polymerase results both from its ability to select the correct nucleotide to incorporate, and from its ability to “proofread” its work.

mutations changes in DNA sequences

Appropriate nucleotide selection depends on base-pairing of the incoming nucleotide with the template strand. At this step, the polymerase makes about one mistake per 1,000 to 10,000 incorporations. Following incorporation, the DNA polymerase has a way of checking to see that the nucleotide pairs with the template strand appropriately (that is, A only pairs with T, C only pairs with G). In the event that it does not, the DNA polymerase has a second enzymatic activity, called a proofreading exonuclease, or a 3'-to-5' exonuclease, that allows it to back up and remove the incorrectly incorporated nucleotide. This ability to proofread reduces the overall error rate to about one error in a million nucleotides incorporated. Other mechanisms detect and remove mismatched base pairs that remain after proofreading and reduce the overall error rate to about one error in a billion.

Features of Replication in Eukaryotic Cells

The steps in DNA replication in eukaryotic cells are very much the same as the steps in bacterial replication discussed above. The differences in bacterial and eukaryotic replication relate to the details of the proteins that function in each step. Although amino acid sequences of eukaryotic and prokaryotic replication proteins have diverged through evolution, their structures and functions are highly conserved. However, the eukaryotic systems are often somewhat more complicated.

For example, bacteria require only a single DNA polymerase, using DNA polymerase III for both leading and lagging strand synthesis, and are able to survive without DNA polymerase I. In contrast, eukaryotes require at least four DNA polymerases, DNA polymerases α , δ , ϵ , and σ . DNA polymerases δ and ϵ both interact with the sliding clamp, and some evidence suggests that one of these polymerases is used for the leading strand and the other for the lagging strand. One required function of DNA polymerase α is the synthesis of the RNA primers for DNA synthesis. The precise role of DNA polymerase σ is not yet known. A second example is removal of the RNA primers on Okazaki fragments. In eukaryotes, primer removal is carried out by RNase H and two other proteins, Fen1 and Dna2, which replace the 5'-to-3' exonuclease provided by the bacterial DNA polymerase I in bacteria.

Replication continues until two approaching forks meet. The tips of linear eukaryotic chromosomes, called **telomeres**, require special replication events. Bacterial chromosomes, which contain circular DNA molecules, do not require these special events.

telomeres chromosome tips

Regulating Replication

DNA replication must be tightly coordinated with cell division, so that extra copies of chromosomes are not created and each daughter cell receives exactly the right number of each chromosome. DNA replication is regulated by



Function(s)	Bacteria	Eukaryotes
single-stranded DNA binding, stimulates DNA polymerase, promotes origin unwinding	SSB (one subunit)	RPA (three subunits)
clamp loader	$\gamma\delta\delta'\tau$ (5 subunits)	RFC (five subunits)
sliding clamp, holds DNA polymerase on DNA	β (two identical subunits)	PCNA (three identical subunits)
replicative DNA polymerase, proofreading exonuclease	DNA polymerase III	DNA polymerase δ (two subunits) DNA polymerase ϵ (four subunits)
DNA primase	DnaG	DNA polymerase α (four subunits)
Okazaki fragment processing	DNA polymerase I RNase DNA ligase H	Dna2 FEN1 RNase H DNA ligase I
DNA helicase	DnaB	?
Swivel ahead of replication fork	ω DNA gyrase	Topoisomerase I Topoisomerase II
Initiator protein	DnaA	Origin Recognition Complex (six subunits)

DNA replication proteins.

controlling the assembly of complexes at replication origins. In bacteria, the accumulation of the initiator protein, *dnaA*, seems to be an important factor in determining when replication begins.

In eukaryotes, DNA replication and cell division are separated by two “gap” cell cycle phases (G_1 and G_2), during which neither DNA replication nor nuclear division occurs. DNA replication occurs during the S (or synthesis) phase, but ORC is thought to bind replication origins throughout the cell cycle. During the G_1 phase of the cell cycle, ORC helps to assemble other replication initiation factors at replication origins to make so-called pre-replicative-complexes (pre-RCs) that are competent to initiate replication during S phase. These other initiation factors include a protein called Cdc6 and a family of six related MCM (“mini-chromosome maintenance”) proteins. The functions of these proteins are not yet known; however, the MCM proteins are currently the best candidate for the eukaryotic replicative helicase, and Cdc6 is necessary for MCM proteins to bind DNA. DNA polymerase α also assembles on origins during this time.

Replication initiation is actually triggered at the beginning of S phase by the phosphorylation (addition of a phosphate group to) of one or more proteins in the pre-RC. The enzymes that phosphorylate proteins in the pre-RC are called protein kinases. Once they become active, they not only trigger replication initiation, but they also prevent the assembly of new pre-RCs. Therefore, replication cannot begin again until cells have completed cell division and entered G_1 phase again. SEE ALSO CELL CYCLE; CHROMOSOME, EUKARYOTIC; CHROMOSOME, PROKARYOTIC; DNA; DNA POLYMERASES; DNA REPAIR; MUTATION; NUCLEASES; NUCLEOTIDE; NUCLEUS; TELOMERE.

Carol S. Newlon

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Reproductive Technology

Successful pregnancy requires ovulation (when an ovary releases an egg into a **fallopian tube**), transport of the egg partway down the fallopian tube, movement of sperm from the vagina to the fallopian tube, penetration by the sperm of the egg's protective layer, and implantation of the fertilized egg in the uterus.

In the United States, infertility is an issue of great concern to many couples of childbearing age. More than 15 percent of all such couples are estimated to be infertile. In a 1995 study by the Centers for Disease Control and Prevention, 10 percent of 10,847 women interviewed, a percentage that represents 6.1 million women of childbearing age nationwide, reported having experienced some problems getting pregnant or carrying a baby to term. Of this group, about half were fertile themselves but had infertile partners. The number of women seeking professional assistance to deal with infertility problems is increasing every year (600,000 in 1968, 1.35 million in 1988, 2.7 million in 1995), and it is reasonable to believe that this trend will continue unabated well into the twenty-first century.

Pregnancy and Infertility

There are many causes of infertility. Abnormal semen causes the infertility problems of about 30 percent of couples seeking treatment. Tubal disease and **endometriosis** in the female partner account for another 30 percent. A female partner's failure to ovulate accounts for 15 percent, and the inability of sperm to penetrate the woman's cervical mucus accounts for another 10 percent. The final 15 percent of couples seeking treatment are infertile for reasons that cannot be diagnosed.

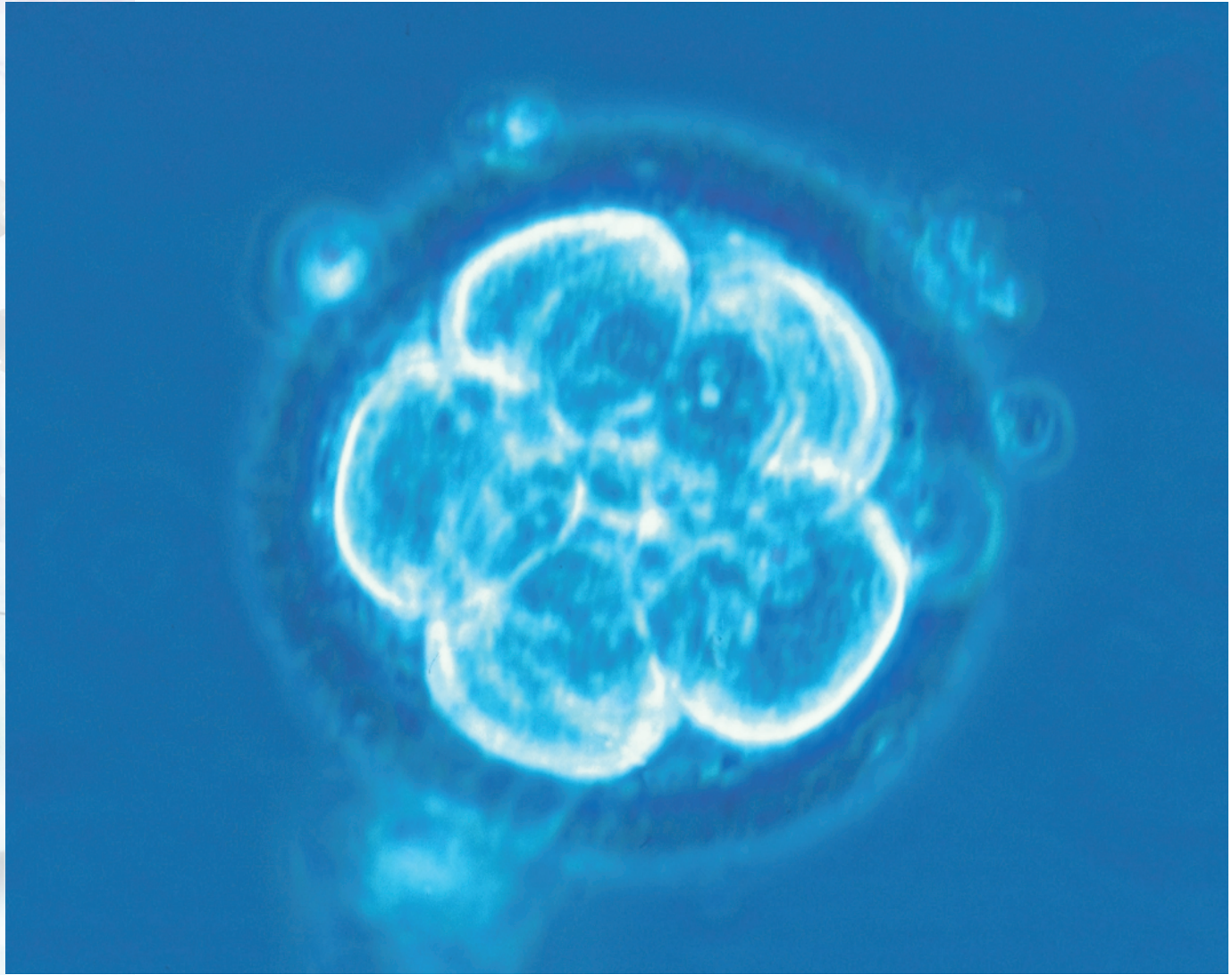
Many couples can be helped to overcome infertility through hormonal or surgical interventions. Women experiencing ovulation disorders may benefit from treatment with oral drugs such as clomiphene citrate, or through the injection of gonadotropins, such as follicle-stimulating hormone, which has about a 75 percent success rate. Women with tubal disease can be helped by various types of reconstructive surgery, but the success rate is only about 33 percent.

However, many infertile couples cannot be helped by such standard methods of treatment. Instead, as a last resort, couples that want children must turn to newer techniques that bypass one or more steps in the usual physiological processes of ovulation, fertilization, and implantation. Commonly

fallopian tube the tube through which eggs pass to the uterus

endometriosis disorder of the endometrium, the lining of the uterus





This eight-cell human embryo is an example of a blastocyst.

in vitro “in glass”; in lab apparatus, rather than within a living organism

cryopreservation use of very cold temperatures to preserve a sample

referred to as “assisted reproductive technology,” these techniques include **in vitro** fertilization (IVF), gamete intrafallopian transfer, zygote intrafallopian transfer, donor insemination, egg donation, embryo **cryopreservation**, intra-cytoplasmic sperm injection, tubal embryo stage transfer, and intrauterine insemination.

In vitro Fertilization

When performed by an experienced practitioner and in an experienced clinic, IVF generally results in pregnancy rates of about 28 percent after one attempt and 51 percent after three. One study has reported the pregnancy rate after six attempts as being 56 percent. Another has reported it as being 66 percent.

Generally, one attempt at IVF is made per menstrual cycle. The IVF process begins when couples are first screened. Clinicians first must rule out infertility in the male partner. If the problem is with the female partner, various courses of treatment may be available. Generally, couples are expected to try to achieve pregnancy for a year after the initial screening

before intervention is attempted. However, if a woman is in her late thirties or older, or if she is experiencing irregular menstruation, a clinical investigation may begin earlier.

Especially in older women, the blood level of follicle-stimulating hormone, a hormone that acts on the ovary to stimulate the maturation of viable eggs, is measured. If the hormone's level is found to be elevated early in a woman's menstrual cycle (after the first week of the new cycle), her ovaries may not be responding to it. In that case, hormonal treatment to stimulate ovulation would be ineffective, and assisted reproductive technology would be unable to help achieve pregnancy. Elevated estrogen levels at day three would also indicate that the ovaries are not responding correctly to estrogen or hormones.

In women whose ovaries are capable of generating viable eggs, the first step in IVF is referred to as "superovulation." To increase the chance of success, the woman's ovaries are stimulated to develop many follicles. Normally, only one or perhaps two follicles develop and are released by an ovary during a single menstrual cycle, which is why usually only one or, on rare occasions, two children are born. In superovulation, a doctor forces multiple follicles to develop so that many **oocytes** can be collected.

To stimulate the ovaries to develop many follicles, the woman undergoes the "long protocol." The action of the pituitary gland is suppressed hormonally, and ten days later the woman is treated with follicle-stimulating hormone. To see how well her ovaries are responding to the hormone, doctors measure estrogen blood levels and observe the ovaries with ultrasound scans. The number and size of the follicles can be visualized. When the doctors judge that the time is right (that is, when the follicle is enlarged to the point that it protrudes above the surface of the ovary), they give the woman human chorionic gonadotropin, wait thirty-six hours, and collect the oocytes from the mature follicles.

In the past, to collect follicles, doctors performed laparoscopy, in which a thin optical tube with a light (called a laproscope) is inserted through a very small incision in the abdominal wall, and the pelvic organs are viewed with fiber optics. Today, the use of a needle guided by ultrasound makes the procedure much faster. The ovary is visualized, mature follicles are located, the needle is inserted, and the follicular fluid that contains the mature oocyte (the unreleased egg) is **aspirated**. The doctors may collect up to eleven oocytes from a single patient.

Viable sperm are collected from the man and washed in a special solution that activates them so they can fertilize the egg. The process of sperm activation is called "capacitation" and normally occurs when sperm are ejaculated and enter the female reproductive tract. Capacitation involves activating enzymes in the sperm's **acrosomal cap**, allowing the sperm head, which contains the sperm's genetic material, to penetrate the outer and inner membranes of the egg (zona pellucida and vitelline membrane). For males with azoospermia, microsurgical or aspiration techniques can directly extract sperm from either the **epididymis** or the testicles. Azoospermia is the most severe form of male infertility, caused by obstructions in the genital tract or by testicular failure.

To allow fertilization to take place, a single egg and about 100,000 sperm are placed together in special culture medium and incubated for about

oocytes egg cells

aspirated removed with a needle and syringe

acrosomal cap tip of sperm cell that contains digestive enzymes for penetrating the egg

epididymis tube above the testes for storage and maturation of sperm



cytoplasm the material in a cell, excluding the nucleus

twenty-four hours. Doctors then use a microscope to see if there are two pronuclei (one from the egg and one from the sperm) in the egg, indicating that fertilization occurred. In some cases, the sperm are unable to penetrate the egg. They may be unable to swim correctly, or they may not have been capacitated successfully. In the past, the only solution was to use sperm from another man. Now, however, sperm can be injected directly into the egg's **cytoplasm**, in a process called microassisted fertilization.

There are three ways used successfully by doctors and researchers to micro-fertilize the egg. The first is "zona drilling," in which a hole is punched into the zona pellucida, letting sperm penetrate the egg. The second method is called "subzonal sperm insertion," in which a sperm is injected directly under the zona pellucida. A third, related method is "intracytoplasmic sperm injection," in which a sperm is injected directly into the egg cytoplasm. The second method is reported to have a higher fertilization success rate (59%) than the third (13%).

The Risks of IVF

In some women, the drugs used to promote superovulation may cause side effects, including mood swings. Some investigators have suggested that procedures used in assisted reproductive technology may not be safe because of the potential for increased ovarian hyperstimulation syndrome and bone loss from the hormone treatments. Ovarian hyperstimulation syndrome occurs at mild levels in 23 percent of women undergoing the treatments, at moderate levels in 3 percent, and at severe levels in 0.1 percent. Complications also may arise as a result of the surgical procedures involved in egg retrieval and embryo transfer. Such complications include pelvic and other infections, which occur in 0.15 percent to 1.2 percent of women, complications from anesthesia, which occur in 0.2 percent, and internal injuries, which occur in 0.38 percent. Although the incidence of such complications is low, every chemical or surgical intervention is associated with risks, and potential patients should be aware of this.

Other concerns regarding the long-term effects of assisted reproductive technology include the increased incidence of spontaneous abortions, which occur in 20 percent of women, and ectopic and heterotopic pregnancies, which occur in 5.5 and 1.2 percent of women, respectively. In heterotopic pregnancies, an embryo is implanted outside the uterus.

There have been conflicting reports as to whether there is a link between the use of fertility drugs and ovarian cancer. Overall, results from many studies do not seem to support such a connection. Some researchers have suggested that the use of the drugs may increase the risk of ovarian cancer later in life, but this is difficult to prove or disprove because the techniques have not been around long enough to assess long-term effects. There have been no reports of any increase in abnormalities in children born using micro-assisted fertilization, though critics have questioned whether the techniques might increase the incidence of such abnormalities.

Embryo Transfer Techniques

Once the egg is fertilized, the two pronuclei fuse to form the early embryo, or "**zygote**." The zygotes are placed in culture **media**, where they undergo cell division, cleaving repeatedly and passing through the two-cell, four-cell,

zygote fertilized egg

media nutrient source

and eight-cell stages. Within seventy-two hours, the zygote develops into the morula, the solid mass of blastomeres formed by the cleavage of the fertilized ovum (egg). After about three to five days in culture, the zygote has become a hollow ball of cells called the **blastocyst**. During normal embryogenesis, it is the blastocyst that is implanted in the endometrial lining of the uterus.

While the embryos are in culture, problems in their development may become apparent. After embryos with evident problems are discarded, one or more cultured embryos are transferred into the uterus, where, it is hoped, one will become implanted and develop into a healthy, full-term baby. Embryo transfer replaces the natural process in which the embryo passes down the fallopian tube and into the uterus, prior to implantation.

Although this transfer is relatively simple and often takes only a few minutes, the rate of successful implantation is low. Usually, two embryos are transferred, but still only one of five women become pregnant. Most doctors who perform IVF procedures adhere to the limit of two embryos per woman to minimize the risk of multiple pregnancies. In most cases, three embryos are transferred in women older than thirty-five. In the United Kingdom, it is illegal for an IVF doctor to transfer more than three embryos at a time into a woman.

A number of factors play a role in whether embryo transfer leads to a baby being born. Success rate is higher if embryo transfer takes place between forty-eight and seventy-two hours after oocyte collection. When more than one embryo is transferred at the same time, the success rate increases, but so does the chance for multiple pregnancies. As noted above, the maximum number transferred should never exceed three. Probably the single most important factor determining whether or not a successful embryo implantation will take place is the donated egg's age. Embryos formed from eggs donated by younger women have a higher implantation success rate than do embryos formed from eggs donated by older women. The age of the host uterus appears to have little or no effect on outcome.

Gamete Intrafallopian Transfer

An alternative to IVF and intrauterine embryo transfer is gamete intrafallopian transfer (GIFT), introduced more than twenty years ago. In this procedure, the egg and sperm are collected as they would be for IVF procedures. However, instead of allowing fertilization to take place in a culture dish, the egg and sperm are transferred surgically into the woman's fallopian tube. This allows fertilization to occur in the fallopian tube, just as occurs in a natural pregnancy. The transfer can only be performed in women with healthy and functional fallopian tubes, and the sperm used for fertilization must be completely normal and capable of swimming.

After the transfer is made, doctors have no way of knowing if normal fertilization actually takes place until an embryo has implanted in the uterine wall. However, the procedure's success rate (35%) is higher than the success rate for IVF. Another related technique is zygote intrafallopian transfer, in which the egg is fertilized in vitro and the zygote is transferred surgically into the fallopian tube. Other techniques include tubal embryo transfer, in which an embryo already undergoing cleavage is transferred.

blastocyst early stage of embryonic development



Intrauterine Insemination

Intrauterine insemination is used when a couple's inability to conceive a child is caused by the sperm's inability to reach the egg. Sperm must move through the uterus and enter the fallopian tube before they can fertilize the egg. Anything that prevents the sperm from making this trip will block conception. Coital or ejaculatory disorders can limit the sperm's travels, sperm antibodies in the female reproductive tract can kill the sperm, and sperm may be unable to penetrate the cervical mucus.

To help the sperm reach the egg, the female is treated with human chorionic gonadotropin to induce multiple ovulation. The number of follicles that are induced is monitored by ultrasound. Washed sperm from the male partner are injected through the cervical opening, into the uterus. The pregnancy rate using this procedure is about 10 percent.

Donor Insemination and Egg Donation

Donor insemination is used when sperm are incapable of fertilizing the egg. Usually this occurs if the male produces very little or no sperm. Sometimes, donor sperm is used when the male partner is the carrier of a genetic disorder that could be transmitted to the baby. Sperm donors should be between ages eighteen and fifty-five, and all should be screened for genetic disorders, such as cystic fibrosis, and for various types of chromosomal abnormalities and infectious disease, including hepatitis, syphilis, cytomegalovirus, and HIV. As with the use of intrauterine insemination, the female partner undergoes ovarian stimulation to maximize the number of follicles released during ovulation. Pregnancy rates resulting from the use of donor insemination are between 32 percent and 50 percent after ten inseminations.

As with donor insemination, egg donation is used when the woman cannot ovulate or is the carrier of a genetic disease. Egg donors must be younger than thirty-five years and must be screened for the same set of conditions as sperm donors. Donors are treated with drugs to stimulate ovulation, after which the eggs are fertilized with the sperm from the male partner and the embryos are transferred to the uterus of the female partner (other procedures can also be used). Growth and development of the embryos then follow the natural processes.

Surrogacy and Cryopreservation

Surrogacy, in which pregnancy occurs in another woman, can supply a couple with an alternative if the woman partner cannot carry the baby to term in her own uterus. In some cases, if the woman cannot supply the egg, sperm from the male partner can be used to inseminate the surrogate mother, who carries the baby to term. Alternatively, if the female partner can produce her own egg, sperm from the male partner can be used to fertilize the egg, and the resulting pre-embryo can be transferred to the uterus of the surrogate mother to grow and develop. Legal controversies resulting from these arrangements have become common in the last few years, so the arrangements should be carefully reviewed by all parties, along with experts in the field, before any final decisions are made.

Frozen sperm and embryos effectively retain their viability for many years. The use of frozen human blastocysts is associated with a 10 percent successful pregnancy rate. Oocyte freezing is much less successful, possibly because oocytes may be genetically damaged or killed in the freezing and thawing. Embryos produced from such cryopreserved eggs have a high incidence of **aneuploidy**, and they are slow to cleave and develop even if they appear to be genetically undamaged. Various research groups are trying to solve this problem.

aneuploidy abnormal chromosome numbers

Age as a Factor

Age must be taken into account when couples are considering assisted reproductive technology. In humans, the age of the oocyte, not the age of the uterus, is the main cause of reproductive failure in IVF and embryo transfer techniques. Embryos formed from older oocytes demonstrate an increased incidence of aneuploidy. In some other species, such as in rabbits, an aging uterus can keep an embryo from being implanted. The use of cryopreservation to circumvent reproductive failure in humans, cattle, and horses has already been successfully employed and is likely to be developed further.

Sperm generated by older men are capable of successfully producing normal embryos. However, as sperm age, they are exponentially more likely to contain new gene **mutations**. Older oocytes, on the other hand, do not appear to be more likely to contain new mutations. Scientists are unsure exactly how age affects oocyte integrity. Oocyte maturation takes place only before birth in the female, so no new oocytes are produced during the entire reproductive life of the female. This is quite different from spermatogenesis, which can continue into old age. Thus, oocytes from older women may be forty or more years old when they are collected and used to form the embryos.

mutations changes in DNA sequences

Oocytes must reach full maturity before they can be ovulated normally and before they can be fertilized, even artificially, to form embryos. If immature oocytes could be artificially forced to mature in vitro, follicles could be taken from the ovaries of dying or dead women, or from cancer patients planning on undergoing chemotherapy treatments, which can damage oocytes. Unlike immature oocytes, immature sperm can effectively be used in fertilization. Additional research is needed in this area of assisted reproductive technology.

Legal, Ethical, and Moral Considerations

The use of these powerful techniques to facilitate reproduction in both humans and animals (the techniques can be used in cattle and pigs, and in the conservation of endangered wildlife) must be balanced against legal, ethical, and moral concerns. For example, would it be permissible to revive extinct animal species? Although a Jurassic Park-like scenario to reanimate extinct dinosaurs is not scientifically credible at this time, what if it became possible to use this technology to form embryos and clone an extinct mammoth, or the passenger pigeon? And what if we can do this for extinct humans? Just because we can develop the capability, would it be acceptable? What are the ethics involved?



Other concerns include questions about how long embryos should remain frozen and who owns frozen embryos not used by the parents. What happens if the parents separate, divorce, or die? What about the legal entanglements involved with surrogacy? Already in the media there have been a number of such cases reported. With the expected increase of these procedures in the future, it is likely that such complex questions will only escalate. Finally, there are basic concerns about helping people sidestep the natural birth process to bring into the world a new human. SEE ALSO CELL CYCLE; CHROMOSOMAL ABERRATIONS; CLONING ORGANISMS; FERTILIZATION; LEGAL ISSUES; MEIOSIS; PRENATAL DIAGNOSIS; REPRODUCTIVE TECHNOLOGY: ETHICAL ISSUES.

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Reproductive Technology: Ethical Issues

Reproductive technology encompasses a range of techniques used to overcome infertility, increase fertility, influence or choose the genetic characteristics of offspring, or alter the characteristics of a population. Each type of reproductive technology brings with it a range of ethical issues. With the accelerated pace of progress in modern medical technology, these issues have been brought squarely into the public arena, where they continue to provoke controversies involving the boundaries of government control, private choice, religious belief, and parental wishes.

Recoiling from Eugenics

Humans' ability to selectively breed desired characteristics into domestic animals and plants, combined with pride and concern for family and for national and ethnic heritage, has led historically to multiple suggestions and experiments aimed at "improving" the human race. These ideas gave rise to the **eugenics** movements in Great Britain and in the United States in the late nineteenth and early twentieth century, preceding and influencing the Nazi "racial hygiene" experiments of the 1930s and 1940s. The horror of these experiments caused an almost universal backlash against eugenic programs that continues to this day.

Nevertheless, from time to time suggestions are made to improve the human gene pool so as to produce people genetically suited for specific tasks, or to improve the general levels of intelligence or other traits that are considered desirable. Numerous ethicists have made strong arguments against such suggestions. The novelist Aldous Huxley decried them in his famous satire *Brave New World*. Paul Ramsey, a Methodist moralist and one of the founders of modern bioethics, argued in his book *Fabricated Man* that any change from natural procreation to mechanized reproductive technology would be harmful to individuals and to society in general.

eugenics movement to "improve" the gene pool by selective breeding



Louise Joy Brown, the first baby conceived by in vitro fertilization was born in England on July 25, 1978.



A child is generally seen as a kind of gift of nature, conceived out of the love and passion of two people. However, as more technology is used in child-birth, there is a concern that the child may be seen increasingly as a commodity whose major purpose is to satisfy the emotional needs of the parents.

Another twenty or thirty years must pass before data will be available on the well-being of children born to couples as a result of reproductive technology, although anecdotal reports are positive. Data about children who have been brought into the world by technology for reasons other than infertility may have to wait another fifty to one hundred years before being analyzed because such children are few in number and difficult to find and study.

The Poles of the Debate

Because reproductive technology encompasses a wide variety of both goals and techniques, there is no single ethical position held by most thinkers considering its ethics. For many, the central, underlying question is when “personhood” begins, in the life of a fertilized **ovum**.

ovum egg

The most conservative stance, opposing all interventions in natural procreation and therefore opposing most reproductive methodologies, is adopted by the Roman Catholic Church and a number of other conservative groups, all of whom invoke some variation of Natural Law ethics. The conservative basis of this nonintervention stance is based on granting full respect to the conceptus from the moment of conception, defined as when a sperm penetrates an ovum.

Some methods used to overcome obstacles to childbirth are permitted by many of these groups. Many would not oppose concentrating a sperm sample from a husband whose sperm count was low, nor would they oppose bringing an ovum around a blocked fallopian tube so sperm could reach the ovum within the mother's reproductive tract. They also would not oppose administering medication to enhance ovulation. Each of these methods involves manipulating the sperm or ovum before conception.

Others who assign the beginning of personhood to later stages of development advocate a greater freedom to utilize reproductive technology depending on the ethical merits of specific situations and on the rights of other individuals who are involved. Within this group, a central question is where to draw the boundary between parental freedoms to choose methods of procreation and to influence the characteristics of their child on the one hand, and, on the other, societal interests in protecting the unborn.

Donor Gametes

One of the oldest and least controversial reproductive technologies is the use of donor sperm to overcome low sperm count on the part of the male or to avoid inheritance of some genetic condition that the male might pass on to his child. Donation is usually anonymous, but some characteristics of the donor are known.

Ethical issues arising in sperm donation include the extent to which parents have the right to choose desirable characteristics in the genetic father of the child, and the right of the child to eventually learn the identity of the father. Each of these issues has precedent in nonassisted reproduction, since prospective parents do choose their mates, and anonymous parenthood occurs in many adoptions. Payment for the sperm sample is generally low enough that the incentive to donate is not thought to be coercive for the donor, and so is not a significant ethical issue.

Donor ova (eggs) are now sometimes used in combination with **in vitro** fertilization technology. Women who are unable to ovulate, or whose ova might transmit a genetic condition they do not want to pass on, use this method. Ovum donation poses risks to the donor. Medication usually is given to cause the release of excess ova, and laparoscopy (in which a needle is inserted through the abdomen) is used to retrieve the ova. Reimbursement is higher and can easily become coercive. Large sums of money have been offered to young women at prestigious colleges, and glamorous potential donors have asked for as much as \$100,000.

In Vitro Fertilization and Surrogacy

In vitro fertilization, even with anonymously donated ovum or sperm, is usually accepted as enabling a couple to experience gestation and create a fam-

in vitro "in glass"; in lab apparatus, rather than within a living organism

ily. However, the in vitro procedure typically creates more embryos than will be implanted. These excess embryos are usually frozen and may remain viable (able to develop normally) for several years. Is it ethical to create embryos that will never be developed? Are the embryos of a couple joint property, which can be divided upon divorce, or should they be considered children, with custody awarded to one parent or the other? What should be done with frozen embryos as they lose their viability?

The ability to fertilize an ovum in a glass dish makes it possible to combine gametes from any man and woman and implant the fertilized ovum into any other woman. Should this be done for other than a husband and wife? For instance, should a reproductive specialist be allowed to refuse to help an unmarried or same-sex couple have a child? Antidiscrimination laws cover some of these issues, but the application of this law to reproductive technology services is still developing.

In vitro fertilization also raises the possibility of employing a woman to be a surrogate mother, acting as the carrier of a child who is expected to be raised by another woman. Women have contracted to render this service in exchange for having all their medical costs covered and in exchange for a stipend to be paid when the baby is delivered. The ethical and legal problems posed by such arrangements have drawn much attention, and many states regulate the boundaries of such agreements.

The birth mother (surrogate) usually signs a contract to allow the adoption of the baby upon birth. However, hormonal changes during pregnancy, especially the increased levels of oxytocin, can produce a variety of behavioral changes, often termed “maternal instinct,” in the pregnant woman. The woman may have been quite willing to surrender the child upon birth, but she may have become unwilling to do so after birth.

In general, the birth mother has the principal right to decide about fulfilling her contract to give over the baby, but the state has stepped in, making contrary decisions when the welfare of the child is at stake. Surrogacy has other potential problems, such as what happens when there are prenatally diagnosed abnormalities and termination decisions must be made or when there are differing standards of prenatal care between the surrogate and the contracting party, and when there is a desire for control of the contracting individual or couple over the surrogate.

Genetic Selection and Medical Motives

Ova and sperm can be genotyped so that particular genes or combinations of genes can be selected. While this is usually very expensive and not yet foolproof, success has been reported in influencing the sex of a child.

Cells from early embryos can be removed for even more precise genotypic determination and selection. Such selection has the potential of being more and more precise. Technology may even become capable of altering traits, as the **genome** becomes better understood. What kind of genetic selection should be allowable in embryos? Should one avoid mental retardation in potential children? Should one select for possibilities of greater intelligence? Is it ethical to choose to have a child only of a particular sex?

Genetic analysis also allows parents to have children whose genes can help others. But should a child be brought into the world simply for the

genome the total genetic material in a cell or organism





stem cell cell capable of differentiating into multiple other cell types

sake of another, or should it be brought in only for its own sake? In a widely reported case in 2000, a couple selected for implantation an embryo that was found to be a perfect bone marrow match for their child, whose own marrow was failing due to Fanconi anemia.

Testing showed that the embryo did not have the recessive disease and that his marrow would not be rejected by his sister, who needed a transplant. The parents attested to their desire for another child, they refused to give birth prematurely for the sick child's sake and they refused to put the infant in any risk by extracting marrow for transplant. They used only the stem cells from the placenta to begin the repopulation of the older daughter's marrow. The process worked! This case highlights the difficult issues involved, and probably represents the best process and outcome possible in such a case. The future may bring more problematic cases as the technology advances to allow treatment of more conditions through tissue or **stem cell** transplantation.

Stem Cells

The existence of stem cells—cells with the potential to develop into a wide variety of other cell types—presents other ethical issues. Placental stem cells may not be able to become every type of cell in the body. In contrast, stem cells derived from developing embryos can. People who believe abortion is ethically acceptable usually have little problem with stem cell collection, because they do not accept the embryo as a person at this stage of gestation.

Most people who believe that personhood begins at conception object to harvesting stem cells from embryos, even from the many thousands of embryos frozen in reproductive clinics that gradually lose their viability as years pass. In 2001 President George W. Bush declared that federal funds could be used for research only on stem cell lines that were already in existence, and that the creation of new lines could not be funded by taxpayer money. Stem cell research is allowed if funded by private sources. Many scientists feel federal money should be allowed to fund stem cell research, citing the enormous potential benefits it can bring. **SEE ALSO** CLONING ORGANISMS; EUGENICS; GENETIC TESTING: LEGAL ISSUES; PRENATAL DIAGNOSIS; PRIVACY; REPRODUCTIVE TECHNOLOGY.

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Restriction Enzymes

Restriction enzymes are bacterial proteins that recognize specific DNA sequences and cut DNA at or near the recognition site. These enzymes are widely used in molecular genetics for analyzing DNA and creating **recombinant DNA** molecules.

Biological Function and Historical Background

Restriction enzymes apparently evolved as a primitive immune system in bacteria. If viruses enter a bacterial cell containing restriction enzymes, the viral DNA is fragmented. Destruction of the viral DNA prevents destruction of the bacterial cell by the virus. The term “restriction” derives from the phenomenon in which bacterial viruses are restricted from replicating in certain strains of bacteria by enzymes that cleave the viral DNA, but leave the bacterial DNA untouched. In bacteria, restriction enzymes form a system with modification enzymes that **methylate** the bacterial DNA. Methylation of DNA at the recognition sequence typically protects the microbe from cleaving its own DNA.

Since the 1970s, restriction enzymes have had a very important role in recombinant DNA techniques, in both the creation and analysis of recombinant DNA molecules. The first restriction enzyme was isolated and characterized in 1968, and over 3,400 restriction enzymes have been discovered since. Of these enzymes, over 540 are currently commercially available.

Nomenclature and Classification

Restriction enzymes are named based on the organism in which they were discovered. For example, the enzyme *Hind* III was isolated from *Haemophilus influenzae*, strain Rd. The first three letters of the name are italicized because they abbreviate the genus and species names of the organism. The fourth letter typically comes from the bacterial strain designation. The Roman numerals are used to identify specific enzymes from bacteria that contain multiple restriction enzymes. Typically, the Roman numeral indicates the order in which restriction enzymes were discovered in a particular strain.

There are three classes of restriction enzymes, labeled types I, II, and III. Type I restriction systems consist of a single enzyme that performs both modification (methylation) and restriction activities. These enzymes recognize specific DNA sequences, but cleave the DNA strand randomly, at least 1,000 **base pairs** (bp) away from the recognition site. Type III restriction systems have separate enzymes for restriction and methylation, but these enzymes share a common subunit. These enzymes recognize specific DNA sequences, but cleave DNA at random sequences approximately twenty-five bp from the recognition sequence. Neither type I nor type III restriction systems have found much application in recombinant DNA techniques.

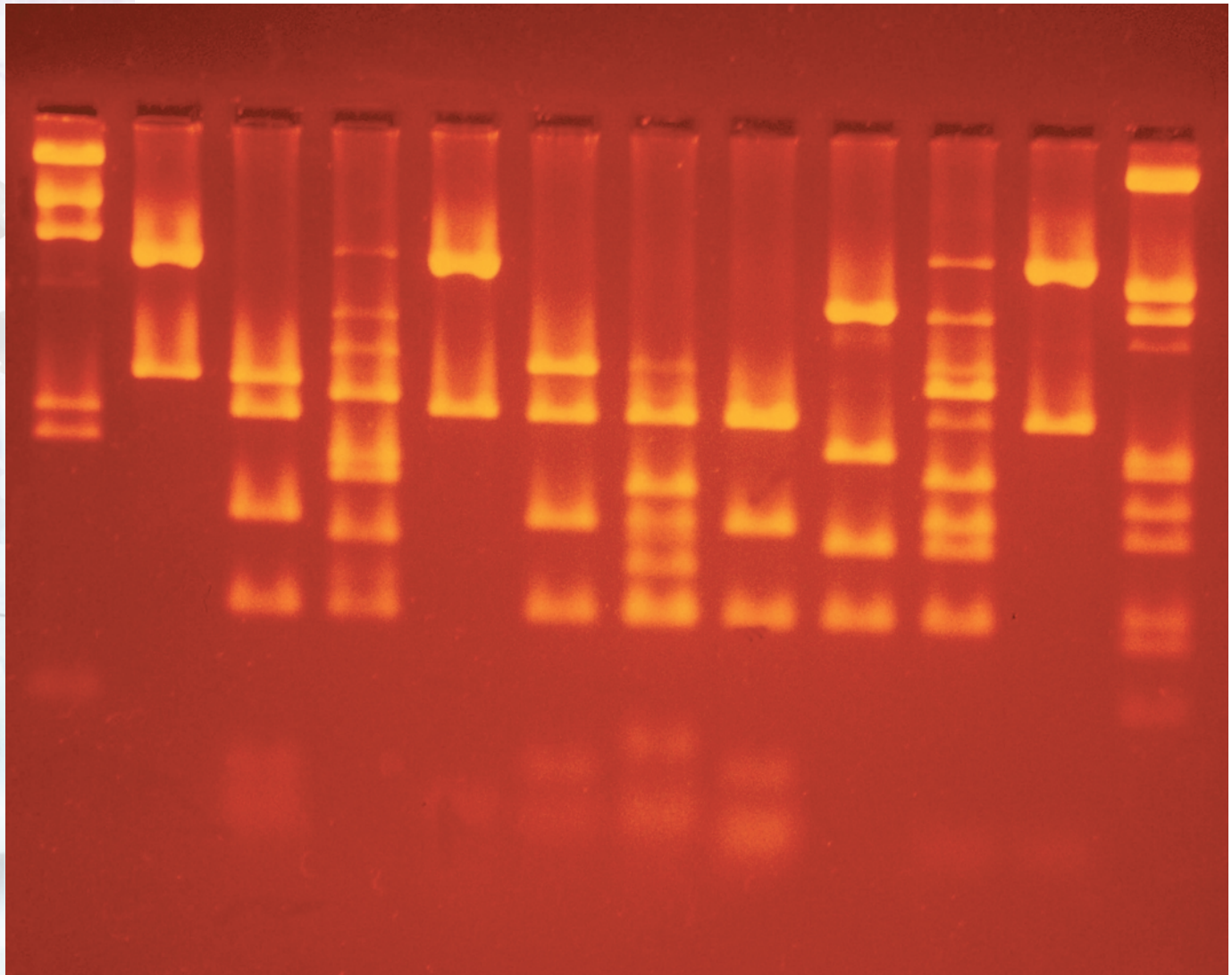
Type II restriction enzymes, in contrast, are heavily used in recombinant DNA techniques. Type II enzymes consist of single, separate proteins for restriction and modification. One enzyme recognizes and cuts DNA, the other enzyme recognizes and methylates the DNA. Type II restriction enzymes cleave the DNA sequence at the same site at which they recognize it. The only exception are type IIs (shifted) restriction enzymes, which cleave

recombinant DNA DNA formed by combining segments of DNA, usually from different types of organisms

methylate add a methyl group

base pairs two nucleotides (either DNA or RNA) linked by weak bonds

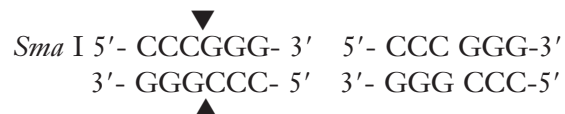




Different restriction enzymes cleave DNA at specific recognition sequences. Cleaving a single piece of DNA with multiple restriction enzymes creates a “DNA fingerprint.” The pattern of fragments can be compared to similar DNA from another source treated with the same enzymes, to determine if the two are identical or different.

DNA on one side of the recognition sequence, within twenty nucleotides of the recognition site. Type II restriction enzymes discovered to date collectively recognize over 200 different DNA sequences.

Type II restriction enzymes can cleave DNA in one of three possible ways. In one case, these enzymes cleave both DNA strands in the middle of a recognition sequence, generating blunt ends. For example:



(The notations 5' and 3' are used to indicate the orientation of a DNA molecule. The numbers 5 and 3 refer to specific carbon atoms in the deoxyribose sugar in DNA.)

These blunt ended fragments can be joined to any other DNA fragment with blunt ends, making these enzymes useful for certain types of DNA cloning experiments.

Pst I 5'- CTGCAG- 3' -CTGCA-3' G-
 3'- GACGTC- 5' G 3'-ACGTC

Bam HI 5'- GGATCC-3' G 5'-GATCC
 3'- CCTAGG-3' CCTAG-5' G

Use of Restriction Enzymes in Biotechnology

Restriction enzymes also have applications in several methods for identifying individuals or strains of a particular species. Pulsed field **gel electrophoresis** is a technique for separating large DNA fragments, typically fragments resulting from digesting a bacterial genome with a rare-cutting restriction enzyme. The reproducible pattern of DNA bands that is pro-

gel electrophoresis
technique for separation
of molecules based on
size and charge

duced can be used to distinguish different strains of bacteria, and help pinpoint if a particular strain was the cause of a widespread disease outbreak, for example.

Restriction fragment length polymorphism (RFLP) analysis has been widely used for identification of individuals (humans and other species). In this technique, genomic DNA is isolated, digested with a restriction enzyme, separated by size in an agarose gel, then transferred to a membrane. The digested DNA on the membrane is allowed to bind to a radioactively or fluorescently labeled probe that targets specific sequences that are bracketed by restriction enzyme sites. The size of these fragments varies in different individuals, generating a “biological bar code” of restriction enzyme-digested DNA fragments, a pattern that is unique to each individual.

Restriction enzymes are likely to remain an important tool in modern genetics. The reproducibility of restriction enzyme digestion has made these enzymes critical components of many important recombinant DNA techniques. SEE ALSO BIOTECHNOLOGY; CLONING GENES; GEL ELECTROPHORESIS; MAPPING; METHYLATION; NUCLEASES; POLYMORPHISMS; RECOMBINANT DNA.

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Retrovirus

Retroviruses are RNA-containing viruses that use the enzyme reverse transcriptase to copy their RNA into the DNA of a host cell. Retroviruses have been isolated from a variety of vertebrate species, including humans, other mammals, reptiles, and fish. The family Retroviridae includes such important human **pathogens** as human immunodeficiency virus (HIV) and human T-lymphotropic virus (HTLV), the causes of AIDS and adult T-cell leukemia respectively. The study of this virus family has led to the discovery of **oncogenes**, resulting in a quantum advance in the field of cancer genetics. Retroviruses are also valuable research tools in molecular biology and gene therapy.

Characteristics

The classification of retroviruses is based on comparisons of the size of the genome and morphologic characteristics (see Table 1). The genomic RNA

pathogens disease-causing organisms

oncogenes genes that cause cancer

Genus	Distinguishing feature	Example	Host	Diseases/pathologies
Alpha-retrovirus	genome <8kb; assembly at cell membrane	avian leukosis virus	birds	malignancies
Beta-retrovirus	intracytoplasmic assembly (B- or D-type)	mouse mammary tumor virus	mice	mammary and ovarian carcinoma; lymphomas
Gamma-retrovirus	genome >8kb; assembly at cell membrane	murine leukemia virus	mice	malignancies
Delta-retrovirus	genomes <9kb; C-type	bovine leukemia virus	cows	malignancies
Epsilon-retrovirus	assembly at cell membrane; hosts: fish	walleye dermal sarcoma virus	fish	solid tumors
Lentivirus	genome >9kb; bar-shaped concentric core	human immunodeficiency virus	humans	immunodeficiency and neurologic disease
Spumavirus	assembly as intracytoplasmic particles	chimpanzee foamy spumavirus	simians	none apparent

of retroviruses is single-stranded and possesses “positive” polarity similar to that found in messenger RNA (mRNA). Virions (virus particles) contain two 5′ (“five prime”), end-linked, identical copies of the genome RNA, and are therefore said to be diploid.

Three genes are universally present in the genomes of retroviruses that are capable of replication, such as murine (mouse) leukemia virus. The *gag* (group antigen) gene encodes proteins that make up the nucleocapsid of the virus as well as a matrix layer, the two of which surround the RNA. The *pol* gene (a type of polymerase) encodes reverse transcriptase, which copies the RNA into DNA, and integrase, which integrates the DNA into the host chromosome. Depending on the species, *pol* can also encode protease, a protein that cleaves the initial multiprotein products of retrovirus translation to make functional proteins. Some retroviruses have incorporated viral oncogene sequences. An example of this is reticuloendotheliosis virus strain T. The genome of complex retroviruses, such as HTLV, can contain several other genes that regulate genome expression or replication and are not present in simple retroviruses.

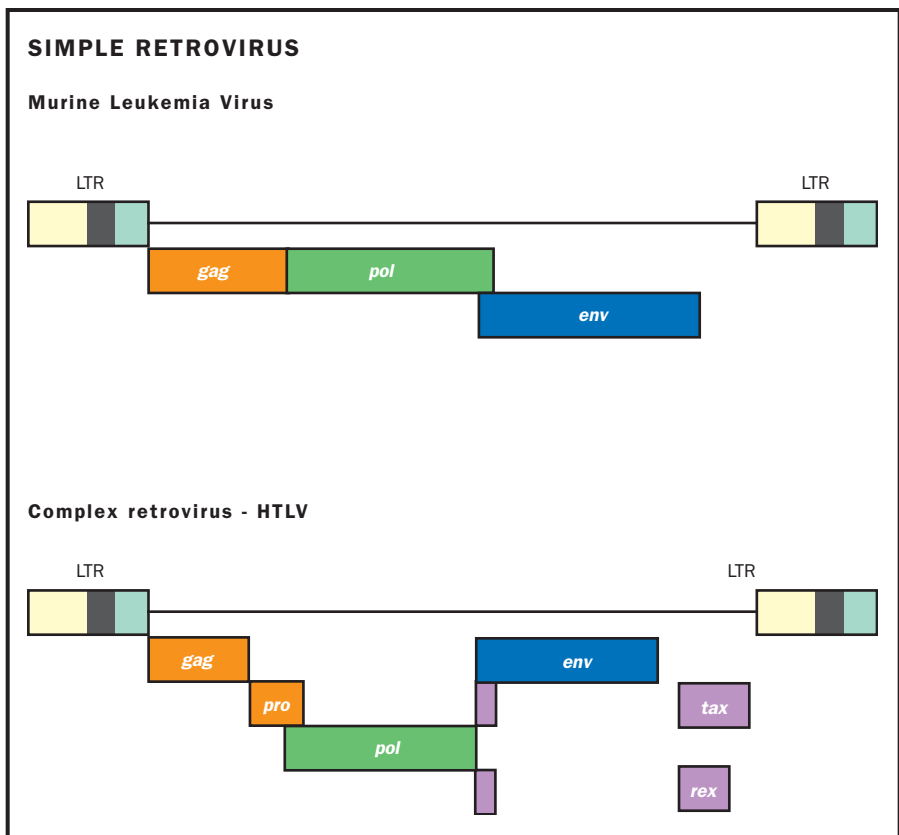
Table 1. Viruses of the family *Retroviridae*.

Reverse Transcriptase

Retroviruses follow the same general steps in their replication cycles that are common to other viruses. The steps that differ from other viruses involve the retroviral reverse transcriptase, an enzyme discovered simultaneously by Howard Temin and David Baltimore in 1970. (Temin and Baltimore were awarded the Nobel Prize for this work in 1975.) Reverse transcriptase converts the single-stranded, positive-polarity RNA genome of retrovirus into double-stranded DNA, thereby reversing the typical flow of genetic information (which is from DNA to mRNA). The DNA copy is transported into the nucleus of the host cell, circularized, and integrated into the host chromosome.

This DNA copy of the retrovirus genome is referred to as the provirus or proviral DNA. The genomes of most vertebrates contain abundant numbers of incomplete and complete proviruses (endogenous retroviruses) that

The genomes of simple retroviruses, such as murine leukemia virus (MLV), contain only three genes, *gag*, *pol*, and *env*. The genomes of complex retroviruses, such as human T-cell lymphotropic virus (HTLV) contain additional genes (e.g., *tax* and *rex*) for regulatory proteins.



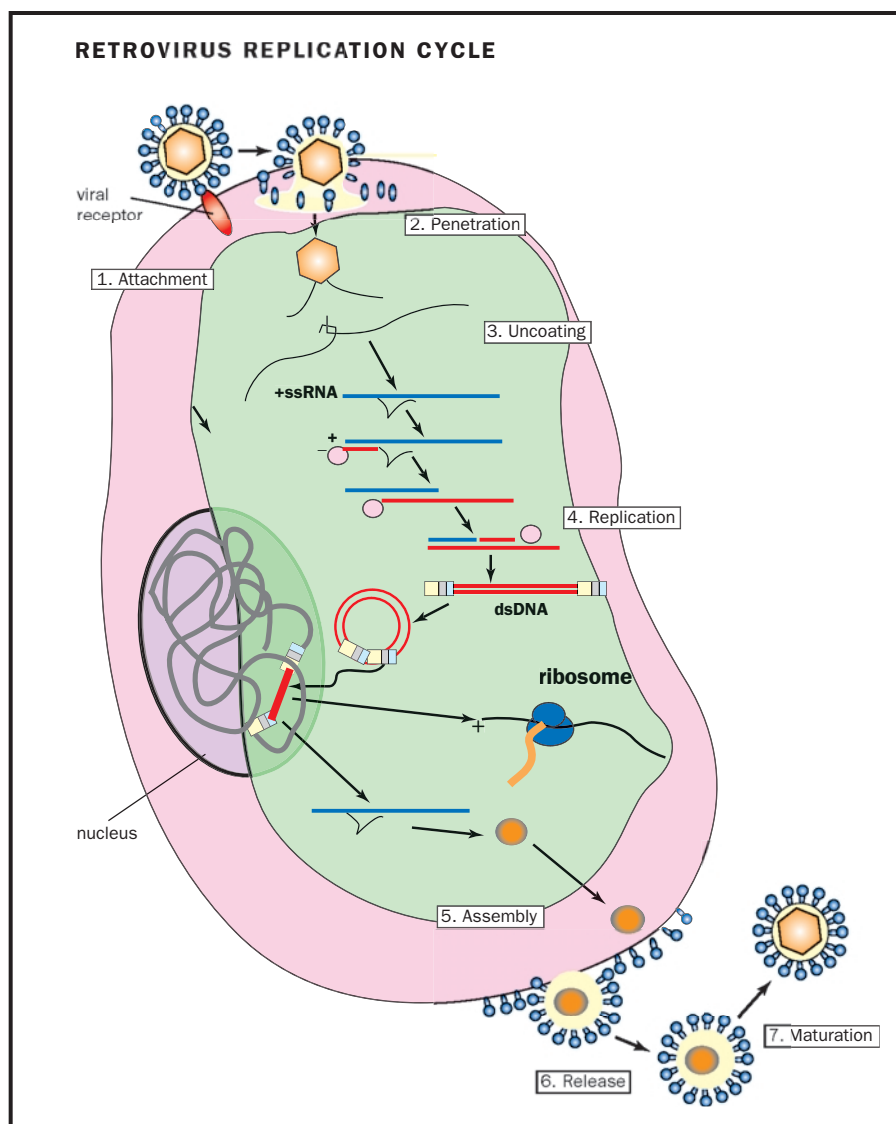
appear to represent remnants of past retroviral infections in germline cells. Proviruses contain structures called long terminal repeats (LTR) at each end. The LTRs contain promoter elements and transcriptional start sites that enable the retroviral genes to be expressed. They can also affect the expression of nearby cellular genes.

Retrovirus Replication Cycle

There are seven steps in the replication cycle of the retrovirus. The first step is attachment, in which the retrovirus uses one of its glycoproteins to bind to one or more specific cell-surface receptors on the host cell. Some retroviruses also employ a secondary receptor, referred to as the co-receptor. Some retroviral receptors and coreceptors have been identified. For example, CD4 and various members of the chemokine receptor family on human T cells (a type of white blood cell) serve as the HIV receptors and coreceptors.

The second and third steps are penetration and uncoating, respectively. Retroviruses penetrate the host cell by direct fusion of the virion envelope with the plasma membrane of the host. Continuation of this fusion process results in the release of the viral capsid directly into the host cell's cytoplasm, where it is partially disrupted.

Step four is replication, which occurs after the retrovirus has undergone partial uncoating. At this stage, the RNA genome is converted by reverse transcriptase into double-stranded DNA. Reverse transcriptase has three enzymatic activities: RNA-directed DNA polymerase makes one



Retrovirus attaches to a host cell membrane, penetrates, and uncoats its genomic RNA. Single-stranded RNA (ssRNA) is used as a template for creation of double-stranded DNA (dsDNA), which is then integrated into the host genome. Once transcribed back into RNA, it codes for viral proteins, is assembled into viral particles, and is released from the cell.

DNA strand, DNA-directed DNA polymerase makes the complementary strand, and RNase H degrades the viral RNA strand. Reverse transcription is primed by a cellular transfer RNA (tRNA) that is packaged into retrovirus virions. It concludes with the synthesis of a double-stranded copy of the retroviral genome that is termed the “provirus,” or proviral DNA.

This proviral DNA is circularized and transported to the host cell’s nucleus, where it is integrated, apparently at random, into the genome by means of the retroviral enzyme called integrase. Following integration, the provirus behaves like a set of cellular genes, while the LTRs function as **promoters** that begin transcription back into mRNA. This transcription is carried out by RNA polymerases in the host cell. Transcription of the proviral DNA is also the means of generating progeny RNA. Viral proteins are made in the cytoplasm of the host cell by cellular ribosomes.

The next step (step five) is termed “assembly,” in which retrovirus capsids are assembled in an immature form at various locations in the host cell.

promoters DNA sequences to which RNA polymerase binds to begin transcription

immunosuppressive describes an agent able to suppress immune system function

autoimmune reaction of the immune system to the body's own tissues

malignancy cancerous tissue

kinases enzymes that add a phosphate group to another molecule, usually a protein

transcription factors proteins that increase the rate of gene transcription

This is followed by an “egress” stage, in which the envelope proteins of retroviruses are acquired by budding from the plasma membrane (cell surface) of the host. Finally, step seven is “maturation.” In this step, the Gag and Pol proteins of the retrovirus are cleaved by the retroviral protease, thus forming the mature and infectious form of the virus.

Consequences of Retroviral Infection

Retroviral infection can result in several different outcomes for the virus and the cell. Retroviruses are capable of inducing **immunosuppressive**, **autoimmune**, and neurological illnesses. Some retroviruses, such as the lentiviruses and the spumaviruses, are capable of directly killing cells. Cytopathic (cell-killing) effects in infected T cells and cells in the brain may account for the profound immune deficiencies and neurological diseases induced by HIV and related lentiviruses.

Retroviruses are also capable of inducing latent infections, in which the virus is dormant, or persistent infections, in which low levels of the virus are continuously produced. These capabilities explain the life-long nature of retroviral infections, and render the diseases induced by these pathogens extremely difficult to treat.

Retroviruses and Cancer

Retroviruses are among several types of viruses that can induce cancer in the host organism. So-called slowly transforming viruses are exemplified by human T-lymphotropic virus (HTLV), which causes leukemia (a type of blood cancer) in humans. These viruses induce **malignancy** by a process called insertional mutagenesis. The initial event is thought to be retroviral integration near, and subsequent activation of, a cellular oncogene (*c-onc*). Examples of *c-onc* include genes for growth factors, protein **kinases**, and **transcription factors**. Harold Varmus and Michael Bishop won the Nobel Prize for physiology or medicine in 1989 for their contributions to the discovery of oncogenes.

When a malignancy is triggered, tumors appear only after a long latent period of months or years, and these tumors are typically clonal in origin. That is, they arise by the rare transformation of a single cell. HTLV-1 is highly prevalent in people living in Japan, the Caribbean, and Africa, areas where approximately one percent of adults are infected. About one to three percent of infected individuals will eventually develop adult T-cell leukemia after an incubation period, which is usually several decades long. HTLV stimulates T-cell proliferation that could favor mutational events leading to cell transformation.

Acutely transforming retroviruses contain a viral oncogene (*v-onc*) and induce polyclonal cancers (that is, many different cancer cells are derived in multiple transforming events) at high efficiency within a short time frame (weeks). The *v-onc* are derived by incorporation and modification (that is, by deletion of introns, mutations, and other such processes) of host-cell oncogenes. The *v-onc* are often expressed in great quantity, due to the highly active viral LTRs. Most acutely transforming retroviruses are replication-defective, because incorporation of the oncogene deletes an essential gene or genes. They therefore require a helper virus to prop-

agate. An exception is Rous sarcoma virus, whose genome retains enough of the structural gene sequences to remain capable of replication. SEE ALSO DNA LIBRARIES; EVOLUTION OF GENES; GENE THERAPY; HIV; ONCOGENES; OVERLAPPING GENES; REVERSE TRANSCRIPTASE; RNA; RNA POLYMERASES; VIRUS.

Robert Garry

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Reverse Transcriptase

Reverse transcriptase is the replication enzyme of retroviruses. Because it **polymerizes** DNA precursors, reverse transcriptase is a DNA polymerase. However, whereas cellular DNA polymerases use DNA as a template for making new DNAs, reverse transcriptase uses the single-stranded RNA in retroviruses as the template for synthesizing viral DNA. This unusual process of making DNA from RNA is called "reverse transcription" because it reverses the flow of genetic information (from DNA to RNA, rather than from RNA to DNA found in transcription). Because reverse transcriptase is essential for retroviruses such as HIV-1 (the virus that causes AIDS), it is the target of many antiretroviral therapeutics. Reverse transcriptase is also a molecular tool used in the cloning of genes and the analysis of gene expression.

Discovery

Retroviruses were originally known as RNA tumor viruses because they have RNA, not DNA, genomes, and because they were the first viruses recognized to cause certain cancers in animals. At the middle of the twentieth century, Howard Temin was interested in understanding how RNA tumor viruses cause cancer. One finding that interested him was the genetic-like stability of the uncontrolled cell growth caused by these viruses. It was known then that certain bacterial viruses, called phages, could integrate their DNA into their hosts' chromosomes and persist as stable genetic elements known as prophages. By analogy, Temin proposed the provirus hypothesis, which suggests that RNA tumor viruses can cause permanent alterations to cells by integrating into host chromosomes. In order for this to occur, Temin suggested that **virion** RNAs were first converted into DNAs, which could then become integrated.

The chemistry of using RNA as a **template** for DNA seemed possible. However, reverse transcription was at odds with the then-popular central dogma of molecular biology, advanced by Francis Crick, which maintained that genetic information flowed unidirectionally from DNA to RNA to protein. RNA tumor viruses were RNA viruses, so it was assumed that their replication involved RNA polymerases, as had been demonstrated for other RNA viruses, and not a DNA polymerase. Because his proposal of a reverse

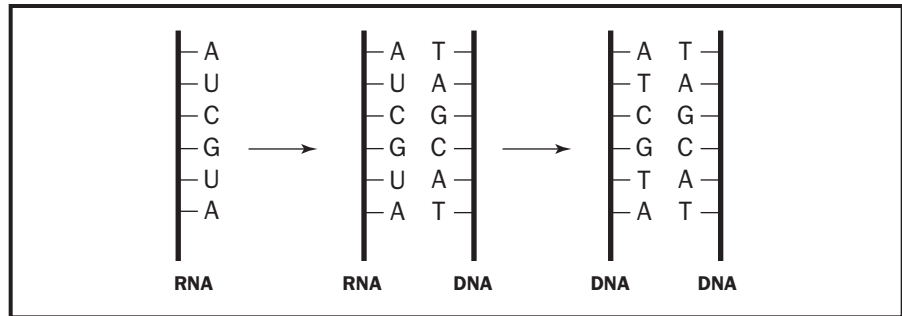
polymerizes links together similar parts to form a polymer

virion virus particle

template a master copy



Reverse transcriptase uses a single-stranded RNA template to create a double-stranded DNA.



flow of genetic information from RNA to DNA seemed heretical, and because the experimental techniques needed to test this idea were not yet developed, Temin and his hypothesis were rebuffed for many years.

The biochemical proof for reverse transcription finally arrived in 1970 when two separate research teams, one led by Temin and the other by David Baltimore, simultaneously discovered the elusive RNA-copying DNA polymerase in purified virions. In 1975 Temin and Baltimore shared the Nobel Prize in physiology or medicine for their discovery of reverse transcriptase.

Laboratory Uses of Reverse Transcriptase

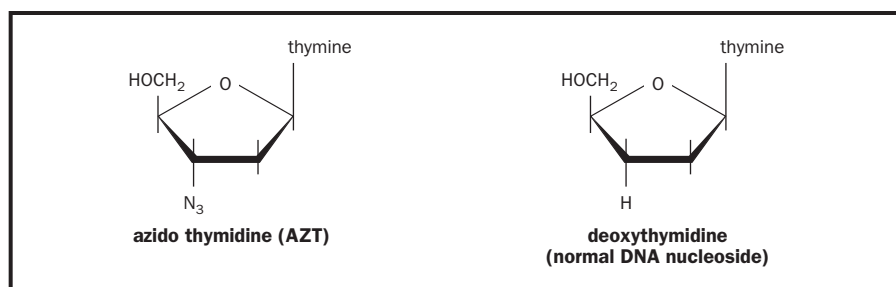
Reverse transcriptase went on to play a critical role in the molecular revolution of the late 1970s and 1980s, especially in the fields of gene discovery and biotechnology. Genes can often be discovered most easily by isolating and analyzing the messenger RNA (mRNA) production in a cell. Reverse transcriptase allowed the synthesis of cDNA, or complementary copies of messenger RNAs. The cDNA can then be expressed in a model organism such as *Escherichia coli*, and the protein it codes for can then be made in abundance. The cloning of cDNA was instrumental to gene discovery in the later part of the twentieth century. Using cDNA copies of genes is necessary when bacteria are used to produce human protein-based pharmaceuticals. This is because bacteria lack the machinery necessary to recognize unspliced genes, but bacteria can use cDNAs to direct the synthesis of human or other higher organism proteins.

Even though the human genome sequence was reported in 2001, copying RNAs with reverse transcriptase remains important. One reason for this is that some human diseases result from **mutations** in genes whose products act to adjust the sequences of RNAs after transcription but before protein synthesis. Thus, even though prototype human sequences are available, it appears likely that molecular diagnostics will include screening cDNA copies of individual people's RNAs. Other uses of cDNA include generating probes to screen microarrays to assess variation in gene expression and regulation.

Reverse Transcriptase and AIDS

Soon after AIDS was recognized in the early 1980s, Luc Montagnier of France and, subsequently, the American Robert Gallo determined that the causative agent was a retrovirus. Like other retroviruses, HIV-1 contains reverse transcriptase and must generate DNA. Differences between reverse

mutations changes in DNA sequences



AZT becomes incorporated into the DNA chain formed by reverse transcriptase, but the N₃ groups block further replication.

transcriptase and cellular DNA polymerases in the sorts of DNA precursors (**nucleosides**) that they can utilize have been exploited to develop drugs that are selectively toxic to HIV-1.

Azidothymidine (AZT) is an example of a nucleoside analog DNA precursor that can serve as a reverse transcriptase “suicide inhibitor,” because AZT incorporation into viral DNA prevents later steps in viral replication. However, the effectiveness of these sorts of drugs is limited by several factors. AZT is occasionally incorporated into cellular DNA, which contributes to the toxicity some patients experience when treated with reverse transcriptase inhibitors. Additionally, reverse transcriptase inhibitor resistance often develops during antiretroviral therapy. This resistance results from reverse transcriptase’s high error rate, which generates a remarkable amount of genetic variation within HIV populations. If some viral genetic variants are less sensitive to antivirals than other variants, the resistant mutants will replicate during antiviral therapy. Despite these complications, reverse transcriptase inhibitors remain important components of the combined antiviral regimen that has dramatically lengthened the lives of many HIV-infected patients since the mid-1990s.

Reverse Transcription and the Human Genome

When reverse transcriptase was first described, it was believed to be a peculiarity of retroviruses. However, researchers now know that reverse transcription also occurs during the replication of the DNA virus hepatitis B, and that RNA-copying DNA polymerases function within human cells. One of these host reverse transcriptases is telomerase, an enzyme that helps maintain chromosome ends.

Other human reverse transcriptases are parts of **endogenous** retroviruses and retroelements, such as those that encoded the majority of the repetitive “junk” DNA in human chromosomes. Many of these retroelements integrated their DNAs into our chromosomes so long ago that they predate human speciation. Because of this, molecular **phylogeneticists** can use sites of retroelement insertions to determine the lineages and ancestral relationships of species. Thus, while retroviruses, in the form of HIV-1, represent one of the newest diseases of humans, the prevalence of other retrovirus-like elements in our genomes demonstrates the long-standing relationship of humans with reverse transcribing elements. **SEE ALSO** DNA MICROARRAYS; DNA POLYMERASES; EVOLUTION OF GENES; HIV; NUCLEOTIDE; RETROVIRUS; TELOMERE; TRANSCRIPTION; TRANSPOSABLE GENETIC ELEMENTS.

nucleosides building blocks of DNA or RNA, composed of a base and a sugar

endogenous derived from inside the organism

phylogeneticists scientists who study the evolutionary development of a species

Alice Telesnitsky